Discrete Structures

Problem Set 4: Generating Functions

Deadline: November 14, 2026, 11:59 p.m.

Instructions

- The marks for each question (or sub-question) are specified alongside it.
- If you use variables that are not defined in the question, state them clearly, and the domain they belong to as well (if applicable).
- Answer questions precisely, maintaining mathematical rigour as much as possible.
- Both digital (written, not typed) and handwritten submissions will be accepted.
- The deadline for this set is November 14, 11:59 p.m.
- Submissions to be done on Moodle with the format `{rollnumber}_DS_A4` (for example `2025113019_DS_A4`).
- Plagiarism and AI use shall be penalised heavily.

Throughout, $\binom{r}{k} = \dfrac{r(r-1)\cdots(r-k+1)}{k!}$ for any real r and integer $k \geq 0$, and $(1+x)^r = \sum_{k\geq 0}\binom{r}{k}x^k$. You may differentiate power series term by term. For $n \geq 0$ let

$$a_n = \binom{2n}{n}, \qquad A(x) = \sum_{n\geq 0} a_n x^n.$$

Question 1 **[18 marks]**

In 1897 the Indiana House of Representatives voted 67–0 for House Bill No. 246, "introducing a new mathematical truth", which implied, among other things, that $\pi = 3.2$. It died in the Senate only because a mathematics professor, C. A. Waldo, happened to be at the statehouse that day. In this question you will do Professor Waldo's job with a recurrence and the binomial theorem.

(a) Show that the sequence (a_n) satisfies the recurrence

$$(n+1)\,a_{n+1} = 2(2n+1)\,a_n \qquad (n \geq 0), \qquad a_0 = 1.$$

Let $c_n = (-4)^n\binom{-1/2}{n}$ be the coefficient of x^n in $(1-4x)^{-1/2}$. Working from the definition of $\binom{r}{k}$, show that (c_n) satisfies the same recurrence with $c_0 = 1$, and conclude that $A(x) = (1-4x)^{-1/2}$. **[6]**

(b) Using (a), show that for $|t| < 1$,

$$\arcsin t = \sum_{n\geq 0} a_n \frac{t^{2n+1}}{4^n(2n+1)}.$$

Hint: you do not need to integrate. Recall that $\dfrac{d}{dt}\arcsin t = \dfrac{1}{\sqrt{1-t^2}}$, and that two functions on $(-1,1)$ with the same derivative that agree at $t = 0$ are equal. Differentiate the right-hand side term by term and compare it with $A(t^2/4)$.

Evaluate this at a suitable value of t. Keep the terms $n = 0, 1, 2$ exactly, and use the recurrence from (a) to show that the remaining terms are bounded by a geometric series. Hence prove that $3.139 < \pi < 3.142$. In particular, $\pi < \frac{22}{7}$, and Indiana's 3.2 is off by more than 0.05. **[7]**

(c) Let $b_n = \sum_{k=0}^{n} 4^{n-k} a_k$ and $B(x) = \sum_{n \geq 0} b_n x^n$. Show that $b_0 = 1$ and $b_n = 4b_{n-1} + a_n$ for $n \geq 1$. Use this recurrence to express $B(x)$ in terms of $A(x)$, and hence show that $B(x) = (1 - 4x)^{-3/2}$. Finally, by differentiating $A(x)$ and using (a), prove that

$$b_n = (2n + 1)\binom{2n}{n} \qquad \text{for every } n \geq 0.$$

[5]

Question 2 **[12 marks]**

Throughout this question, $[x^n]\, F(x)$ denotes the coefficient of x^n in $F(x)$. You may use, without proof, that

$$\frac{1}{(1-x)^k} = \sum_{n \geq 0} \binom{n+k-1}{k-1} x^n \qquad (k \geq 1).$$

Use generating functions in every part; an answer obtained only by another counting method (for example stars and bars, or direct listing) receives no credit for that part, although you may use listing to check your answer.

Kavish's team enters a hackathon, where every project is scored by 4 judges. Each judge gives an integer score from 0 to 10, and the project's total is the sum of the four scores. A *score-sheet* is the ordered list (s_1, s_2, s_3, s_4) of the four scores.

(a) Explain why the number of score-sheets with total n is $[x^n]\, S(x)$, where

$$S(x) = \left(\frac{1 - x^{11}}{1 - x}\right)^4.$$

Hence find the number of score-sheets with total exactly 20. **[4]**

(b) Kavish says: "$S(x) = \left(\frac{1-x^{11}}{1-x}\right)^4$ becomes $\frac{0}{0}$ at $x = 1$, so $S(1)$ is meaningless. Besides, we were told never to substitute values into a generating function." Karthik replies that Kavish is only half right. Explain which part of Kavish's claim is right and which part is wrong, and then use $S(1)$ to find the total number of score-sheets. **[3]**

(c) Show that $x^{40} S(1/x) = S(x)$, and deduce that, for every n, the number of score-sheets with total n equals the number with total $40 - n$. Using this and part (b), find the number of score-sheets with total at least 21. **[5]**

Question 3 **[16 marks]**

On 16 October 1843, William Rowan Hamilton was walking with his wife into Dublin when, just as they reached Brougham Bridge over the Royal Canal, the rule

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$$

came to him, and he cut it into a stone of the bridge. In the 1920s his numbers turned up inside every electron, proton and quark: the quaternions of norm 1 form exactly the group $SU(2)$

that physicists use to describe *spin*. This is why turning an electron through 360° multiplies its quantum state by -1, and only a 720° turn brings it back; neutron interference experiments have observed this sign flip directly. Let Q be the ring of real quaternions from the lectures, whose elements are $q = \alpha_0 + \alpha_1\mathbf{i} + \alpha_2\mathbf{j} + \alpha_3\mathbf{k}$ with $\alpha_t \in \mathbb{R}$. For such q, define its *conjugate* and *norm* by

$$\bar{q} = \alpha_0 - \alpha_1\mathbf{i} - \alpha_2\mathbf{j} - \alpha_3\mathbf{k}, \qquad N(q) = \alpha_0^2 + \alpha_1^2 + \alpha_2^2 + \alpha_3^2.$$

(a) Show that $q\bar{q} = \bar{q}q = N(q)$ for every $q \in Q$. Deduce that Q is a division ring, that Q is not a field, and that if $ab = 0$ in Q then $a = 0$ or $b = 0$. **[6]**

(b) Over a field, a polynomial of degree 2 has at most two roots. Show that, in contrast, $q^2 = -1$ in Q if and only if $\alpha_0 = 0$ and $\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 1$, so Hamilton did not find one new square root of -1 but a whole sphere of them. Bazinga! Then point out exactly which step of the usual argument "$q^2 + 1 = (q - \mathbf{i})(q + \mathbf{i}) = 0$, so $q = \mathbf{i}$ or $q = -\mathbf{i}$" fails in Q, and why. **[5]**

(c) Prove that $N(pq) = N(p)\,N(q)$ for all $p, q \in Q$. (*Hint:* in the matrix form of Q from the lectures, $N(q)$ is a determinant.) Deduce that a product of two sums of four integer squares is again a sum of four integer squares, and use quaternions to write $105 = 7 \cdot 15$ as one, starting from $7 = 2^2 + 1^2 + 1^2 + 1^2$ and $15 = 3^2 + 2^2 + 1^2 + 1^2$.

Now let $L = \{\alpha_0 + \alpha_1\mathbf{i} + \alpha_2\mathbf{j} + \alpha_3\mathbf{k} : \alpha_t \in \mathbb{Z}\}$, which you may assume is a subring of Q with unity 1. Find all the units of L, and show that L is not a division ring. **[5]**

Question 4 **[16 marks]**

An element e of a ring R is called an *idempotent* if $e^2 = e$. Throughout this question, a ring homomorphism is a map φ with $\varphi(a + b) = \varphi(a) + \varphi(b)$ and $\varphi(ab) = \varphi(a)\,\varphi(b)$ for all a, b. In particular, $\varphi(1) = 1$ is *not* required.

(a) Let R be a commutative ring with unity. Show that if e is an idempotent of R with $e \neq 0$ and $e \neq 1$, then both e and $1 - e$ are zero divisors.
Then find all idempotents of $\mathbb{Z}_{42}$, and justify that your list is complete. **[6]**

(b) Show that the number of ring homomorphisms $\mathbb{Z}_m \to \mathbb{Z}_{42}$ is 2^k, where k is the number of primes among $2, 3, 7$ that divide m.
Hence list all ring homomorphisms $\mathbb{Z}_{28} \to \mathbb{Z}_{42}$, and find the number of ring homomorphisms $\mathbb{Z}_{2027} \to \mathbb{Z}_{42}$. **[10]**

Question 5 **[15 marks]**

For a commutative ring R with unity and $f \in R$, let $(f) = \{fg : g \in R\}$ be the ideal generated by f; the quotient ring $R/(f)$ has the cosets $g + (f)$ as its elements. In $\mathbb{Z}_2[x]$, write

$$K = \mathbb{Z}_2[x]/(x^2 + x + 1), \qquad \omega = x + (x^2 + x + 1).$$

(a) Show that every element of K can be written as $a + b\omega$ with $a, b \in \mathbb{Z}_2$, so K has exactly four elements. Using $\omega^2 = \omega + 1$, write down the multiplication table of the three nonzero elements of K, and conclude that K is a field. **[6]**

(b) Show that $\mathbb{Z}_2[x]/(x^2 + 1)$ is *not* a field by finding a nonzero element whose square is 0. Which factorisation of $x^2 + 1$ over $\mathbb{Z}_2$ is responsible? **[4]**

(c) The polynomial $y^2 + y + 1$ has no root in $\mathbb{Z}_2$. Find all of its roots in K, and write $y^2 + y + 1$ as a product of two linear factors in $K[y]$. **[5]**

Question 6 **[23 marks]**

For centuries $\sqrt{-1}$ was treated with suspicion; Descartes called such numbers "imaginary" as an insult. In 1847 Cauchy proposed a way to make them respectable: do not invent a new number at all, just compute with ordinary real polynomials in x and agree to treat $x^2 + 1$ as 0. In this question you will check that Cauchy was right, and see what goes wrong when one sign changes. Let $I = (x^2 + 1)$ and $J = (x^2 - 1)$ in $\mathbb{R}[x]$.

(a) Using the division algorithm, show that every coset $f + I$ of $\mathbb{R}[x]/I$ contains exactly one polynomial of the form $a + bx$ with $a, b \in \mathbb{R}$. Write the product $(a + bx + I)(c + dx + I)$ in this form. **[5]**

(b) Let $\varphi : \mathbb{R}[x] \to \mathbb{C}$ be given by $\varphi(f) = f(i)$. Show that φ is a surjective ring homomorphism and, using (a), that $\ker \varphi = I$. Conclude that $\mathbb{R}[x]/I \cong \mathbb{C}$, and hence that I is a maximal ideal of $\mathbb{R}[x]$. **[8]**

(c) Assume the analogue of (a) for J. Show that $a + bx + J \mapsto (a + b,\ a - b)$ is a ring isomorphism $\mathbb{R}[x]/J \to \mathbb{R} \times \mathbb{R}$. Use it to find all idempotents of $\mathbb{R}[x]/J$, each written as $a + bx + J$, and conclude that J is not a maximal ideal. **[6]**

(d) Show that $\mathbb{R}[x]/(x^2)$ is not a field, and that it is not isomorphic to $\mathbb{R} \times \mathbb{R}$ either. (*Hint:* look for a nonzero element whose square is zero.) **[4]**